

HW5: Offline Reinforcement Learning

SAC+BC, Implicit Q-Learning (IQL), and Flow Q-Learning (FQL)

1. SAC+BC

SAC+BC combines Q-value maximization with behavioral-cloning regularization. The coefficient α controls how strongly the learned policy is constrained toward actions in the offline dataset.

1.1 Best SAC+BC agents

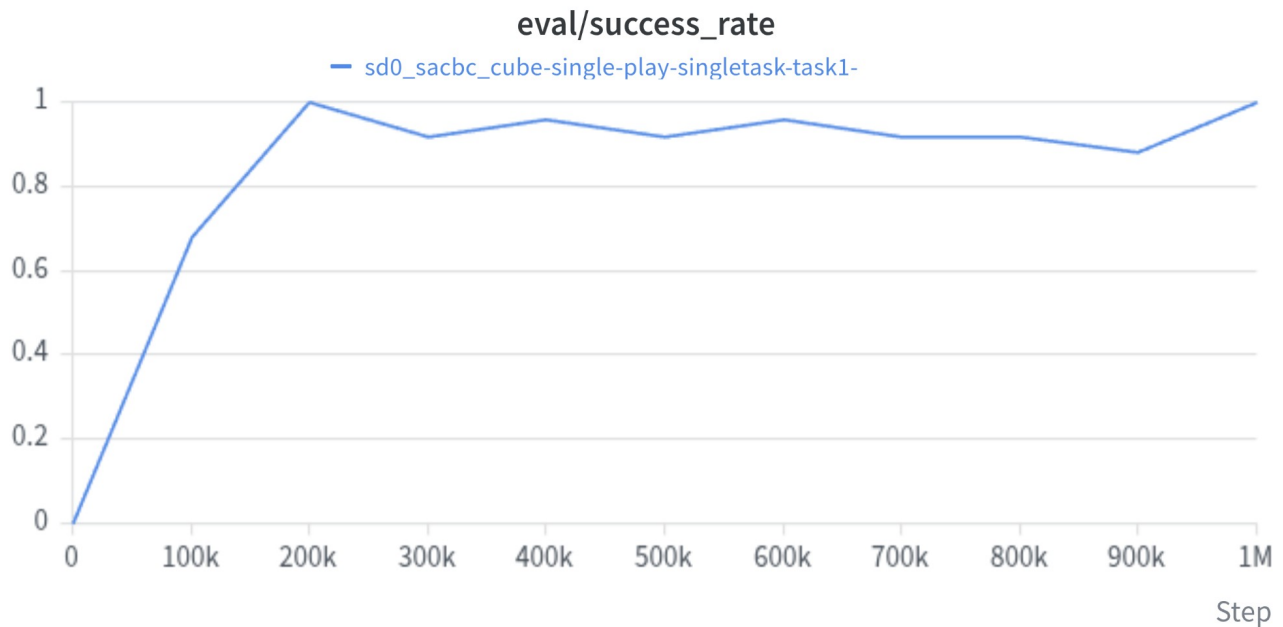


Figure 1. Best SAC+BC agent on cube-single.

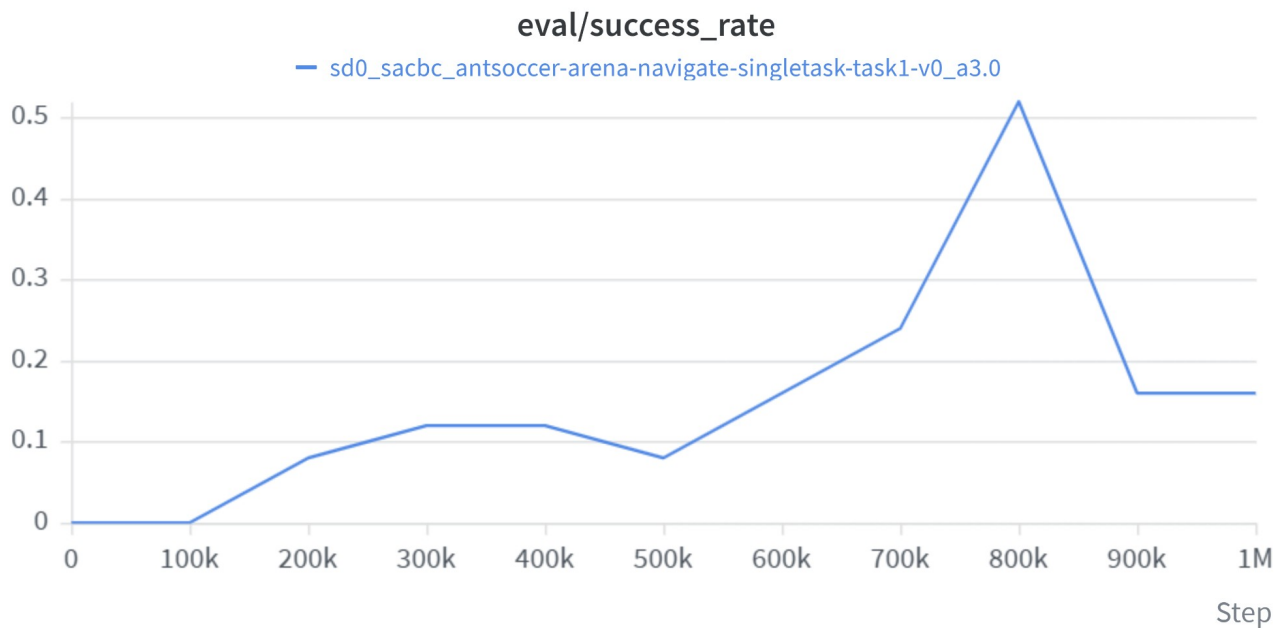


Figure 2. Best SAC+BC agent on antsoccer-arena.

How well does the best SAC+BC agent perform on cube-single?

The best SAC+BC agent performs very well on cube-single. The success rate reaches 100% at around 200K training steps and remains mostly above 90% afterward, reaching 100% again at 1M steps. This comfortably exceeds the required 75% final success rate.

How well does SAC+BC perform on antsoccer-arena?

SAC+BC achieves a substantial non-zero success rate on the more challenging antsoccer-arena task. Performance gradually improves and peaks at approximately 52% around 800K steps, although it fluctuates afterward.

1.2 Effect of α on cube-single

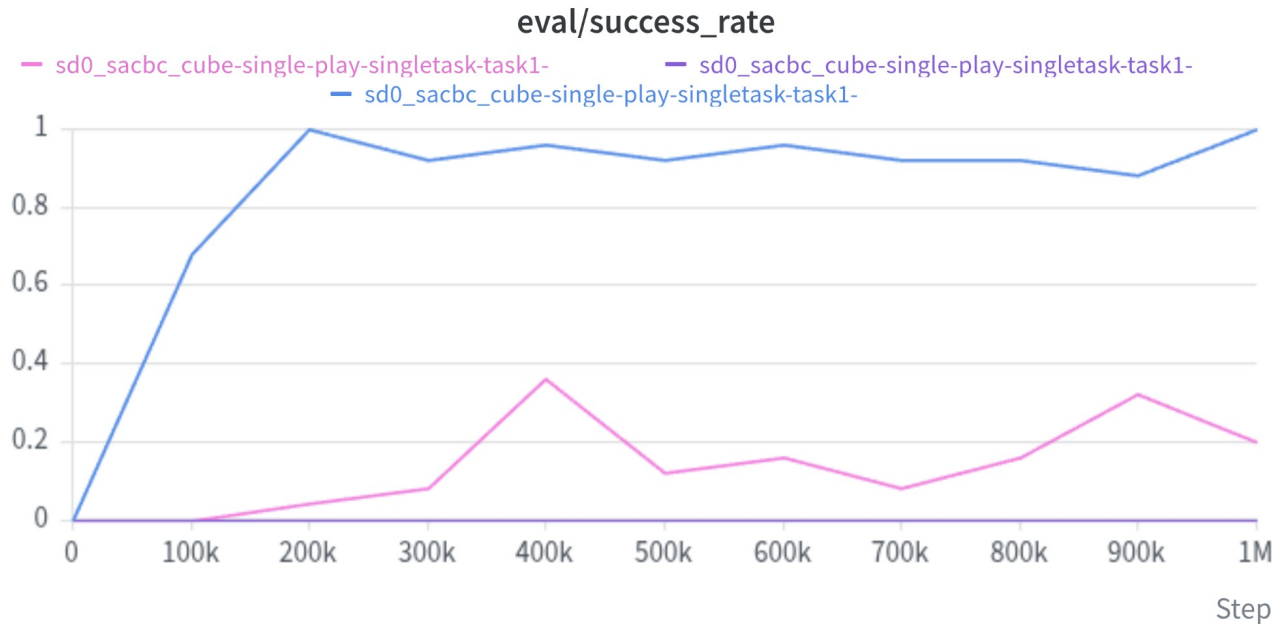


Figure 3. SAC+BC training curves for three different α values on cube-single.

How does the choice of α affect performance on cube-single?

The choice of α has a large effect on SAC+BC. The best setting achieves roughly 90-100% success, while another reaches only about 35% at its best and the poorest setting remains near 0%. Thus, SAC+BC is highly sensitive to the balance between Q maximization and behavioral-cloning regularization.

2. Implicit Q-Learning (IQL)

IQL learns a value function using expectile regression and extracts a policy using advantage-weighted regression. With the expectile fixed in this assignment, α controls the strength of advantage weighting.

2.1 Best IQL agents

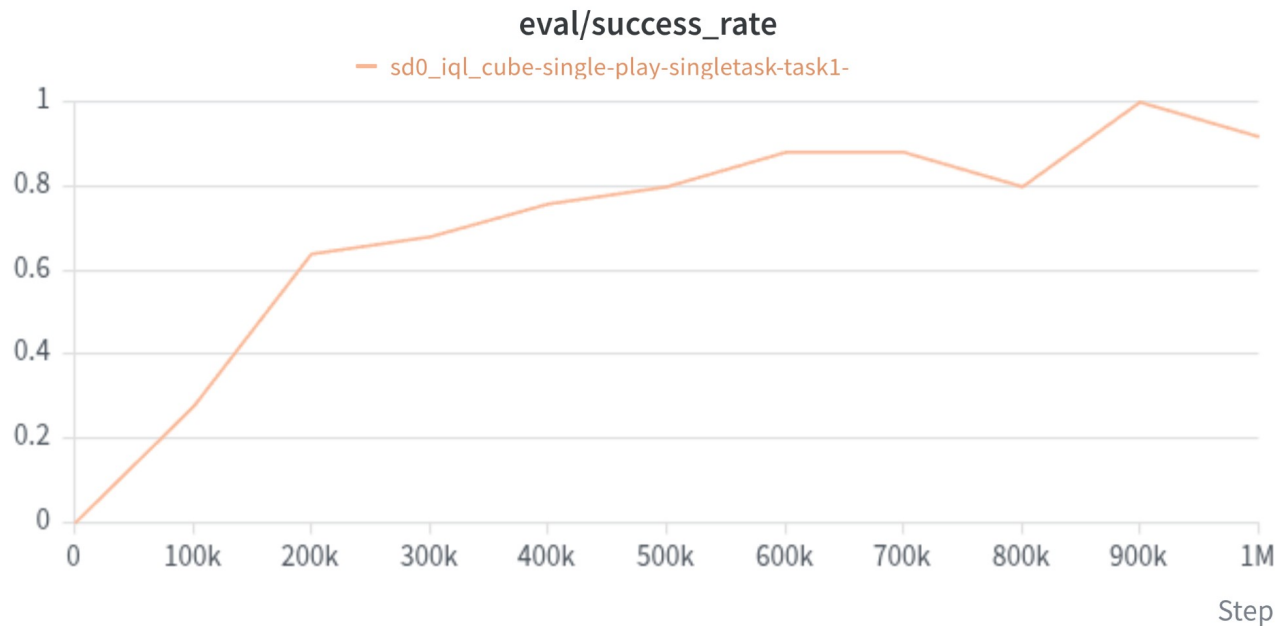


Figure 4. Best IQL agent on cube-single.

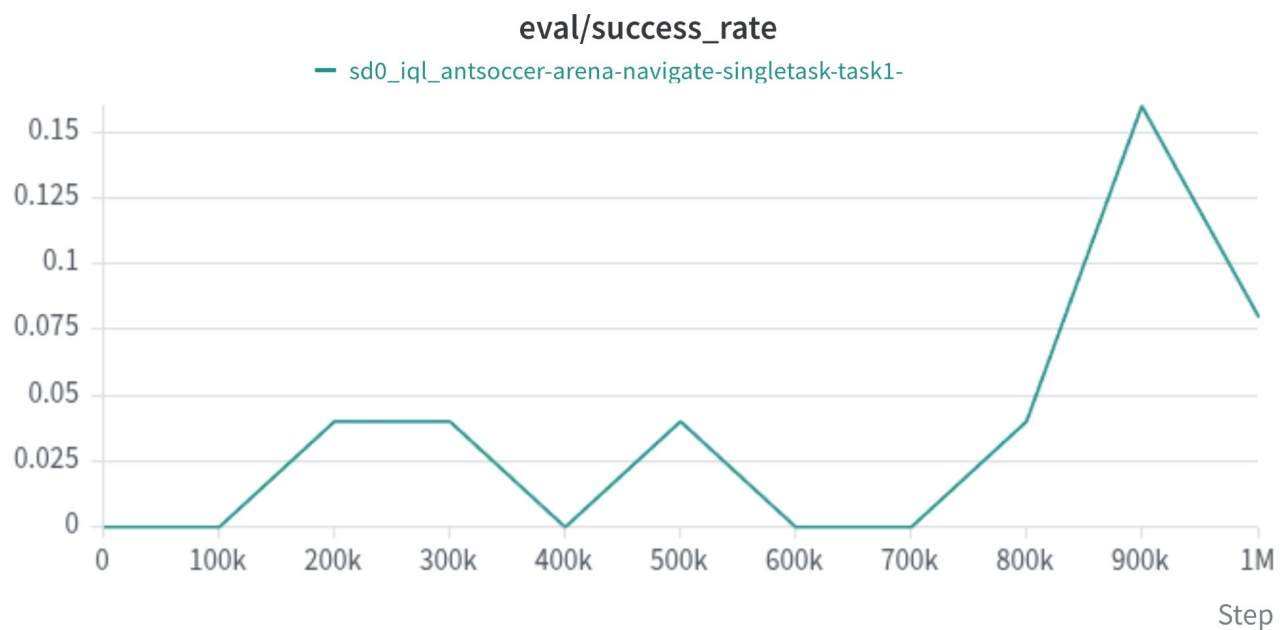


Figure 5. Best IQL agent on antsoccer-arena.

How well does the best IQL agent perform on cube-single?

The best IQL agent performs well on cube-single. Its success rate increases throughout training, reaches 100% around 900K steps, and finishes at approximately 90% at 1M steps, exceeding the required 60% final success rate.

How well does IQL perform on antsoccer-arena?

IQL achieves non-zero but relatively low performance on antsoccer-arena. Its success rate peaks at approximately 16% around 900K steps and finishes near 8% at 1M steps.

2.2 Effect of α and comparison with SAC+BC

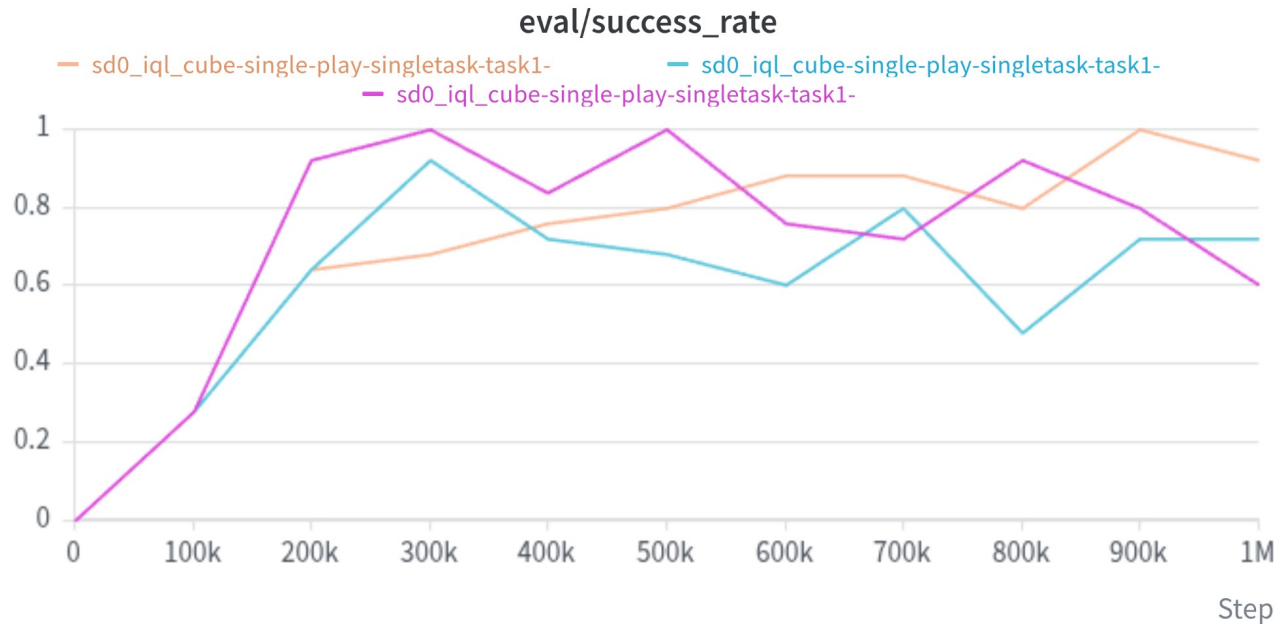


Figure 6. IQL training curves for three different α values on cube-single.

How does the choice of α affect IQL performance on cube-single?

All three tested α values achieve relatively high success rates at some point in training, although their learning curves and final performance differ. Compared with SAC+BC, IQL is therefore relatively robust to α on cube-single.

Compare SAC+BC and IQL in terms of best performance and sensitivity to α .

SAC+BC achieves the stronger best performance on antsoccer-arena, with a peak success rate of about 52% compared with about 16% for IQL. On cube-single, both can achieve high success, but IQL is less sensitive to α because all three tested settings learn useful policies, whereas SAC+BC ranges from near-zero to almost perfect performance.

3. Flow Q-Learning (FQL)

FQL combines an expressive behavioral flow policy with a one-step policy. The one-step policy maximizes Q-values while being distilled toward the behavioral flow policy, avoiding backpropagation through the full ODE integration.

3.1 Best FQL agents

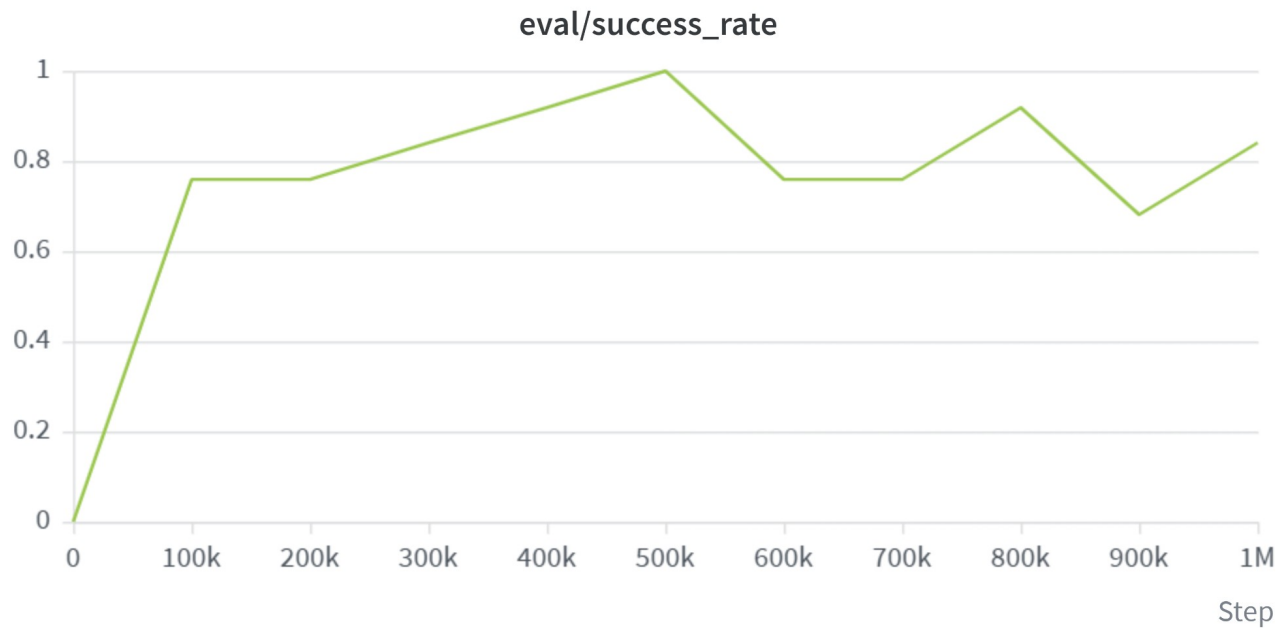


Figure 7. Best FQL agent on cube-single.

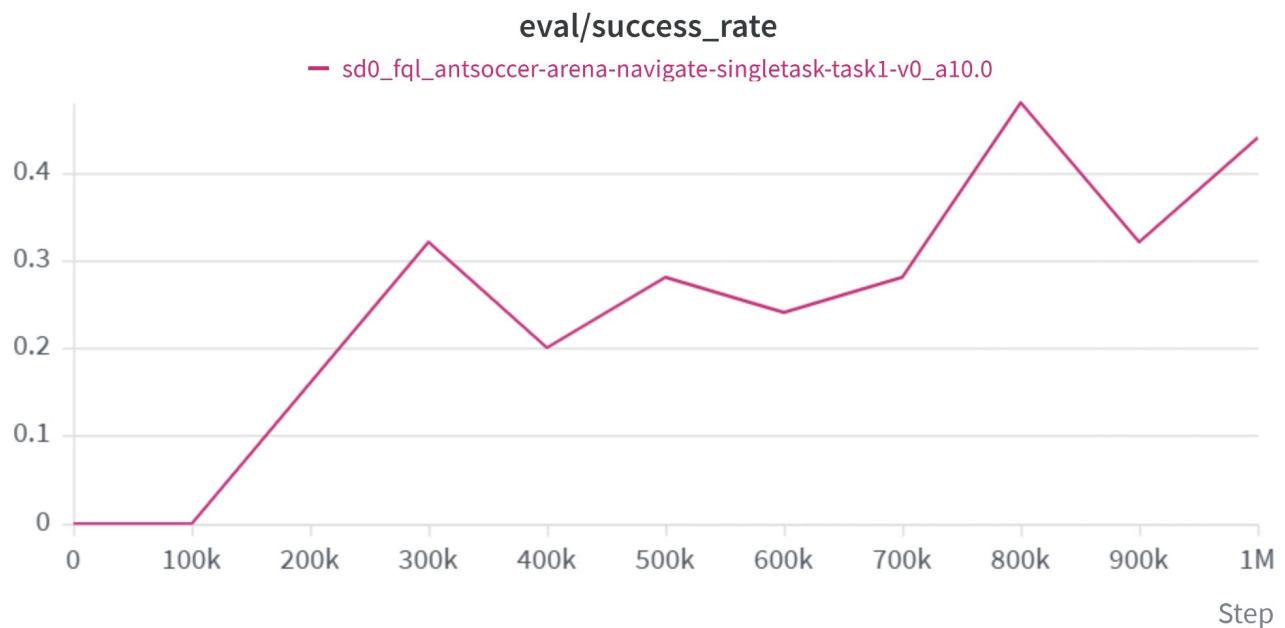


Figure 8. Best FQL agent on antsoccer-arena.

How well does the best FQL agent perform on cube-single?

The best FQL agent performs strongly on cube-single. The success rate reaches approximately 76% at 100K steps, reaches 100% around 500K steps, and finishes at approximately 84% at 1M steps. Therefore, the run satisfies the assignment requirement of a final success rate above 80%.

How well does FQL perform on antsoccer-arena?

FQL achieves strong performance on antsoccer-arena. The success rate reaches approximately 32% at 300K steps, peaks at about 48% around 800K steps, and finishes near 44% at 1M steps. Its best success rate therefore exceeds the required 30% threshold.

What can be observed from the FQL training results?

FQL learns successful policies on both tasks. It reaches near-perfect performance on cube-single while also obtaining meaningful success on the substantially harder antsoccer-arena task. Although the curves fluctuate, both tasks retain strong non-zero performance at the end of training.

4. Overall Results

Method	cube-single	antsoccer-arena	Observation
SAC+BC	Final 100%	Peak ~52%	Highest arena peak; highly sensitive to α
IQL	Final ~90%	Peak ~16%	Robust to α , but weaker on arena
FQL	Final ~84%	Peak ~48%, final ~44%	Meets both FQL performance targets

Overall, all three methods achieve high success on cube-single. SAC+BC obtains the highest observed peak on antsoccer-arena, closely followed by FQL, while IQL is substantially weaker on that task. The α sweeps also show that SAC+BC is much more sensitive to α than IQL.